Battery Monitoring System - High-Level Design

Document version 1.0

Status: released for review

1 Introduction

This High-Level Design (HLD) describes the software architecture for monitoring of cell voltages and pack temperature and estimation of the state of charge for a 48 V battery pack. It lists the software components, their ports and interfaces, the signals exchanged between them and the main functional flows.

The document is the reference for architects, developers, integrators and test engineers. Names of components, interfaces and signals are written exactly as they appear in the software architecture model.

1.1 Scope and assumptions

The design covers the application layer only. Basic software, bootloader and network management are described in separate documents. All components run in the same ECU and communicate through the runtime environment (RTE) using sender-receiver ports.

It is assumed that every signal has exactly one producer. Each signal is refreshed in the cycle time given in the signal list.

1.2 Terms

Term	Meaning
SWC	Software component, the smallest deployable unit of application software.
Port	Connection point of a component. Provided ports send data, required ports receive data.
Interface	Typed description of the data elements exchanged over a port.
Signal	Named data item with one producer and one or more consumers.
RTE	Runtime environment that connects the components.

2 Architecture Overview

The system consists of 4 software components. Each one has a clearly defined responsibility, and the dependencies between components are kept as small as possible. The following sections give the details of every component.

Components that provide data are placed close to the hardware abstraction, while controller components sit above them and make the decisions.

3 Software Components

3.1 Swc_BatMonMgr

Top level manager. It collects voltage, temperature and state of charge and decides about warnings and derating requests.

Swc_BatMonMgr has 3 port(s). The ports and the interfaces they use are listed in the table below.

Port	Direction	Interface
Rp_CellVolt	required	If_CellVoltage
Rp_PackTemp	required	If_PackTemp
Pp_SocValue	provided	If_SocValue

Dependencies: Swc_CellVoltAcq, Swc_TempAcq.

Error handling: invalid or missing input data leads to a safe default value and the component sets its own status to invalid. The status is available to the diagnostic manager.

3.2 Swc_CellVoltAcq

Measures the voltage of each cell through the monitoring chip every 10 ms.

Swc_CellVoltAcq has 1 port(s). The ports and the interfaces they use are listed in the table below.

Port	Direction	Interface
Pp_CellVolt	provided	If_CellVoltage

Dependencies: none.

Error handling: invalid or missing input data leads to a safe default value and the component sets its own status to invalid. The status is available to the diagnostic manager.

3.3 Swc_TempAcq

Reads the pack temperature sensors and removes implausible values.

Swc_TempAcq has 1 port(s). The ports and the interfaces they use are listed in the table below.

Port	Direction	Interface
Pp_PackTemp	provided	If_PackTemp

Dependencies: none.

Error handling: invalid or missing input data leads to a safe default value and the component sets its own status to invalid. The status is available to the diagnostic manager.

3.4 Swc_SocEstimator

Estimates state of charge from the cell voltage using a lookup table and coulomb counting.

Swc_SocEstimator has 2 port(s). The ports and the interfaces they use are listed in the table below.

Port	Direction	Interface
Rp_CellVolt	required	If_CellVoltage
Pp_Soc	provided	If_SocValue

Dependencies: Swc_BatMonMgr.

Error handling: invalid or missing input data leads to a safe default value and the component sets its own status to invalid. The status is available to the diagnostic manager.

4 Interfaces

All interfaces are sender-receiver interfaces. The data elements of each interface are listed below.

Interface	Type	Data elements
If_CellVoltage	Sender-Receiver	CellVoltMv, CellIndex
If_PackTemp	Sender-Receiver	TempDegC, SensorId
If_SocValue	Sender-Receiver	SocPercent, SocValid
If_BalancingCmd	Sender-Receiver	BalanceMask, BalanceTime

5 Signal List

Producer and consumers of every signal are given in the table. A dash means that no consumer is defined.

Signal	Producer	Consumers	Cycle
Sig_SocPercent	Swc_SocEstimator	Swc_BatMonMgr	20 ms
Sig_PackTempC	Swc_TempAcq	Swc_BatMonMgr, Swc_SocEstimator	20 ms
Sig_CellMinMv	Swc_CellVoltAcq	-	20 ms
Sig_BalancingActive	-	Swc_BatMonMgr	20 ms

6 Functional Flows

6.1 Low voltage warning

Step 1: Swc_CellVoltAcq publishes the cell voltages.

Step 2: Swc_BatMonMgr compares the minimum cell voltage with the warning threshold.

Step 3: A warning is raised if the value is below the threshold for 500 ms.

7 Revision History

Version	Date	Change
1.0	2026-04-01	Initial release